



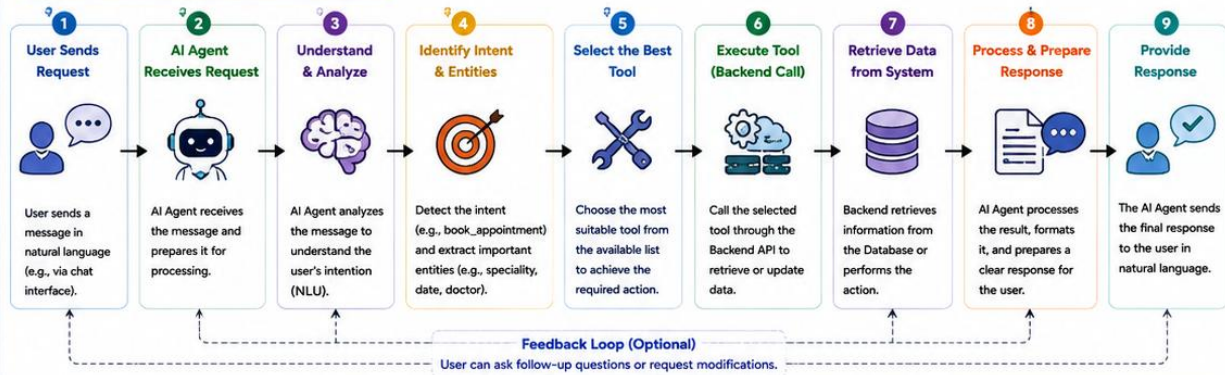
Hospital AI Agent – Workflow

AI-Powered Smart Hospital Management System

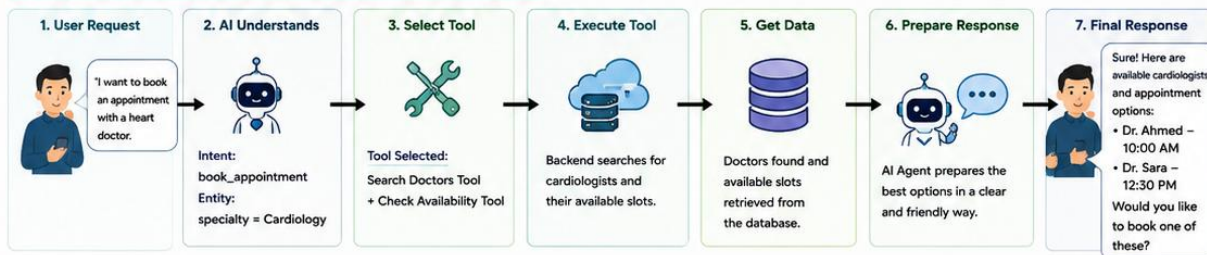


The AI Agent acts as an intelligent assistant inside the system.
It understands user requests, decides the best action, uses the appropriate tools,
retrieves information from the system, and provides a helpful response.

1. AI AGENT WORKFLOW – GENERAL FLOW



2. EXAMPLE SCENARIO – BOOK APPOINTMENT WITH CARDIOLOGIST



3. TOOLS USED BY THE AI AGENT

Tool Name	Purpose	Input	Output	Used By
Search Doctor Tool	Search doctors by speciality or name	speciality / doctor_name	List of doctors	Patient, Receptionist, AI Agent
Check Availability Tool	Check available appointment slots	doctor_id, date	List of available slots	Patient, Receptionist, AI Agent
Book Appointment Tool	Create a new appointment	patient_id, doctor_id, date, time	Appointment confirmation	Patient, Receptionist, AI Agent
Get Patient Info Tool	Retrieve patient information	patient_id	Patient details	Doctor, AI Agent (authorized)
Summarize History Tool	Summarize patient medical history	patient_id	Summary text	Doctor, AI Agent (authorized)
Generate Report Tool	Generate statistics and reports	report_type, date_range	Report file / data	Admin, AI Agent
Send Reminder Tool	Send reminders and notifications	user_id, message, time	Notification status	System, AI Agent

4. KEY POINTS

- ✓ The AI Agent never makes medical diagnoses.
- ✓ It supports users by providing information and suggestions.
- ✓ All actions are performed using controlled tools and authorized APIs.
- ✓ Data access is based on role and permissions.
- ✓ Sensitive patient data is protected.
- ✓ The AI Agent learns from interactions and gets smarter over time.

5. SAFEGUARDS & RULES

- 6 Role-based access control (RBAC).
- 6 Only authorized data is accessible.
- 6 All tools have validation and constraints.
- 6 Important actions are logged.
- 6 Patient privacy and security are always maintained.

Flow Legend - - - - - Information Flow → Process Flow ↔ Two-way Interaction

The AI Agent is an intelligent layer on top of the system. It helps users get things done faster, easier, and smarter.